

Component propagation of the sun-dual resolvent inclusion

Partial answer to an open question in arXiv:2203.12765

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Status. `candidate_partial_result_likely_valid_needs_human_review`.

Abstract

Kruse and Schwenninger ask whether the restricted adjoint resolvent of a bi-continuous semigroup always maps the sun dual into the domain of its generator at every point of the original resolvent set. We prove that the set of points where this inclusion holds is a union of connected components of the resolvent set. It contains the component containing the right half-plane beyond the growth bound. Thus the answer is affirmative whenever the resolvent set is connected, and any counterexample must live in a different, unseeded resolvent component.

1 Source question and proof intuition

Let $(X, \|\cdot\|, \tau)$ be a sequentially complete d-consistent Saks space, let $(T(t))_{t \geq 0}$ be a bi-continuous semigroup with generator $(A, D(A))$, and use the source paper's sun dual $X^\bullet$ and generator $A^\bullet$. For $\lambda \in \rho(A)$, write

$$R(\lambda, A)^\bullet = R(\lambda, A)'|_{X^\bullet}$$

whenever the restriction is viewed as an operator on the sun dual.

The second open question after Proposition 4.8 in [1] asks whether

$$R(\lambda, A)^\bullet X^\bullet \subset D(A^\bullet) \quad (\lambda \in \rho(A)) \tag{1}$$

always holds. The source paragraph is reproduced in Figure 1.

The key observation is local-to-global. At one point where (1) holds, Theorem 4.7(a) of the source identifies the restricted adjoint resolvent with the genuine resolvent of $A^\bullet$. A Neumann series propagates this identity locally. Conversely, norm-resolvent continuity and closedness of $A^\bullet$ preserve the property under limits. The good set is therefore relatively open and closed.

2 The component theorem

Define

$$\Omega = \{\lambda \in \rho(A) : R(\lambda, A)^\bullet X^\bullet \subset D(A^\bullet)\}.$$

Theorem 1 (Component propagation). *The set Ω is relatively open and relatively closed in $\rho(A)$. Consequently, Ω is a union of connected components of $\rho(A)$. It contains the component containing $\{\lambda \in \mathbb{C} : \operatorname{Re} \lambda > \omega_0(T)\}$.*

Proof. Fix $\lambda_0 \in \Omega$. By Theorem 4.7(a) of [1],

$$\lambda_0 \in \rho(A^\bullet), \quad R(\lambda_0, A)'|_{X^\bullet} = R(\lambda_0, A^\bullet). \quad (2)$$

Put $R_0 = R(\lambda_0, A)$ and $S_0 = R(\lambda_0, A^\bullet)$. If $|\lambda - \lambda_0| \|R_0\| < 1$, the Neumann formula gives

$$R(\lambda, A) = R_0 [I + (\lambda - \lambda_0) R_0]^{-1}.$$

Since R'_0 leaves $X^\bullet$ invariant and restricts there to S_0 , taking adjoints and restricting the norm-convergent series yields

$$R(\lambda, A)'|_{X^\bullet} = S_0 [I + (\lambda - \lambda_0) S_0]^{-1} = R(\lambda, A^\bullet).$$

The final operator maps $X^\bullet$ into $D(A^\bullet)$, so a neighbourhood of λ_0 lies in Ω . Thus Ω is relatively open.

Now let $\lambda_n \in \Omega$ and $\lambda_n \rightarrow \lambda \in \rho(A)$. For $x^\bullet \in X^\bullet$, set

$$y_n = R(\lambda_n, A)' x^\bullet, \quad y = R(\lambda, A)' x^\bullet.$$

Norm-resolvent continuity gives $y_n \rightarrow y$ in X' . Each $y_n \in D(A^\bullet) \subset X^\bullet$, and $X^\bullet$ is norm closed in X' , hence $y \in X^\bullet$. Again by Theorem 4.7(a),

$$A^\bullet y_n = \lambda_n y_n - x^\bullet \longrightarrow \lambda y - x^\bullet \quad \text{in } X^\bullet.$$

The operator $A^\bullet$ is closed on $X^\bullet$, so $y \in D(A^\bullet)$. This holds for every $x^\bullet$, proving $\lambda \in \Omega$. Therefore Ω is relatively closed.

A relatively clopen subset is a union of connected components. Proposition 4.8(a) of [1] gives (1) for every $\operatorname{Re} \lambda > \omega_0(T)$. That half-plane is connected, so its entire component lies in Ω . $\square$

Corollary 1 (Connected resolvent set). *If $\rho(A)$ is connected, then (1) holds for every $\lambda \in \rho(A)$.*

Proof. Theorem 1 makes the nonempty set Ω clopen in the connected space $\rho(A)$, so $\Omega = \rho(A)$. $\square$

3 Upgrade attempts and remaining obstruction

The theorem reduces a possible failure to a connected component of $\rho(A)$ disjoint from the right half-plane. Such a component may be a bounded spectral hole, although boundedness is not asserted in general. Four focused upgrades were pursued.

1. A Riesz–Runge functional-calculus argument propagates through the seeded component but cannot cross spectrum separating it from another component; the missing pole is precisely the obstruction.
2. Abstract invariant-subspace examples can make a restriction acquire extra spectrum (the bilateral/unilateral shift mechanism), but no realization was found in the special norming sun-dual subspace of a d-consistent Saks space.
3. Standard shift, multiplication, and translation semigroups examined in this setting have matching resolvent behaviour and do not produce a failure.

4. An ordinal endpoint-projection example initially appeared to answer the separate continuity-necessity question. It is excluded by d-consistency: its endpoint functional is sequentially mixed-continuous but not mixed-continuous, while condition (ii) applied to the bounded weak-null sequence formed by its scalar multiples would force mixed continuity.

Thus the unresolved mechanism is extra spectrum of the restricted adjoint on an unseeded resolvent component, under all the d-consistency and sun-dual constraints. Further local resolvent manipulations do not supply a seed there.

Novelty and review. The four cheap run indexes and exact-phrase/keyword searches found no prior answer to the open question or statement of Theorem 1. Because the proof is short, it may nevertheless be unindexed folklore; novelty confidence is moderate. Expert review should check the restriction of the Neumann series and the closed-graph limit.

Limitations. This is not a full answer in the disconnected case. It also does not settle whether mixed-topology continuity of $R(\lambda, A)$ is necessary for the domain inclusion.

References

- [1] K. Kruse and F. L. Schwenninger, *Sun dual theory for bi-continuous semigroups*, arXiv:2203.12765 (2022; revised 2023).

Part (a) shows that the continuity of $R(\lambda, A): (X, \gamma) \rightarrow (X, \gamma)$ need not be a necessary condition for $R(\lambda, A)^{\bullet} X^{\bullet} \subset D(A^{\bullet})$ for all $\operatorname{Re} \lambda > \omega_0(T)$. This is an open question. Another open question is whether one actually has $R(\lambda, A)^{\bullet} X^{\bullet} \subset D(A^{\bullet})$ for all $\lambda \in \rho(A)$ in general. The answer is affirmative if τ coincides with the $\|\cdot\|$ -topology. Because then γ also coincides with the $\|\cdot\|$ -topology, which gives that $R(\lambda, A): (X, \gamma) \rightarrow (X, \gamma)$ is continuous for all $\lambda \in \rho(A)$. Therefore Proposition 4.8 (b) and Theorem 4.7 imply [54, Theorem 1.4.2, p. 10] (see also [27, Theorem 14.3.3, p. 425]).

Figure 1: Source page 18: the two open questions following Proposition 4.8.